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HISTORY · Consolidation of British rule: policies and administration

UNDERSTAND IT FIRST

What is it? After Plassey and Buxar the Company was a **ruler**, but a company of traders could not run a country. Between **1773 and 1858** the British built a **system of control**: **laws from Parliament** (to control the Company), **Governors-General** (to win and organise territory), **land revenue systems** (to collect money), and **four pillars of administration**: the **civil service, army, police and judiciary**. Learn it as **Acts → Governors-General → Land revenue → Four pillars → Andhra link → Impact**.

Acts of Parliament

1. Acts that controlled the Company

Act	Main points
Regulating Act, 1773	First step by Parliament, after the Company's misrule and bankruptcy. **Governor of Bengal became Governor-General** (**Warren Hastings** first) with a **council of four**; **Bombay and Madras** came under Bengal; **Supreme Court at Calcutta (1774)**, Chief Justice **Elijah Impey**; servants banned from **private trade and gifts**.
Pitt's India Act, 1784	**Board of Control** (6 commissioners) for **political affairs** while the **Court of Directors** kept **commerce and patronage**: **dual control**. Governor-General's council cut to **three**. Company's lands called **'British possessions in India'**.
Charter Act, 1793	Company's trade monopoly **renewed for 20 years**; Governor-General got power over the other presidencies.
Charter Act, 1813	**Monopoly ended** except in **tea and the China trade**; Indian trade open to all British merchants; **₹1 lakh a year for education**; **Christian missionaries** allowed; first claim of the Crown's **sovereignty** over British India.
Charter Act, 1833	Trade monopoly **ended completely**; Company became a **purely administrative body**; **Governor-General of Bengal became Governor-General of India** (**Bentinck** first); **all legislative power** in the Governor-General's council; a **Law Member** added (**Macaulay**); Indians not to be barred from jobs by **religion, birth or colour** (promise in principle).
Charter Act, 1853	**Separate Legislative Council** for law-making; **open competitive examination** for the Civil Service (patronage ended); Law Member became a full member; **last Charter Act**. **1858**: the Crown took over (a later topic).

2. Governors-General and their work

Governor-General	Key policies
Warren Hastings (1772–85)	Ended **dual government (1772)**; **Collectors** appointed; **Sadr Diwani Adalat** and **Sadr Nizamat Adalat** at Calcutta; **Rohilla War (1774)**; **First Maratha War (Treaty of Salbai, 1782)**; **Second Mysore War**; **Asiatic Society (1784)** founded by William Jones; **impeached** in England (1788–95) and **acquitted** (Burke led the charge).
Lord Cornwallis (1786–93)	**Permanent Settlement (1793)**; **Cornwallis Code (1793)**: **separation of revenue and justice** (the Collector lost judicial power), district courts, appeal courts; **police** reorganised with **daroga** at each station; **covenanted civil service** with high pay, no private trade (the **'father of civil service'**); Treaty of Seringapatam (1792).
Lord Wellesley (1798–1805)	**Subsidiary Alliance** (Day 13); defeated Tipu (1799); **Fort William College (1800)** to train civil servants; aggressive expansion.
Lord Hastings (1813–23)	**Anglo-Nepal War** (1814–16, **Treaty of Sugauli**); crushed **Pindaris**; **Third Maratha War (1817–18)**, **Peshwa removed**; Ryotwari spread; **press restrictions eased (1818)**; policy of **paramountcy**.
Lord William Bentinck (1828–35)	**Abolished sati (Regulation XVII, 1829)** with Raja Ram Mohan Roy's support; suppressed **thuggee** (Sleeman); **English education** (Macaulay's Minute, 1835); **Calcutta Medical College (1835)**; Indians in lower judicial posts; **Mahalwari** refinements (1833); reduced army **batta**; **Mysore** placed under direct rule (1831).
Lord Dalhousie (1848–56)	**Doctrine of Lapse**: a dependent ruler without a natural heir could not adopt one without the Company's consent, so the state **lapsed**: **Satara (1848), Jaitpur and Sambalpur (1849), Baghat, Udaipur, Jhansi (1853), Nagpur (1854)**; **Awadh (1856)** was taken for **'misgovernment'**, not lapse. Won the **Punjab (1849)** after the Second Sikh War, **Lower Burma (1852)**, Sikkim (1850). **Modernisation: railways** (first train **Bombay–Thane, 16 April 1853**), **telegraph (1851–54)**, **uniform postage (1854)**, **Public Works Department**, **Ganga Canal**, **Wood's Despatch (1854)**: the 'Magna Carta of English education', leading to **universities at Calcutta, Bombay and Madras (1857)**.

3. Land revenue systems

UNDERSTAND IT FIRST

Why this matters: land revenue was the Company's **biggest income** (about **half** of it). Rates were **very high** and the systems changed who owned land, so they shaped **rural life, debt and famine**. The three systems differ in **who paid the tax**.

Three land revenue systems

System	Details

Permanent Settlement (Zamindari), 1793	**Cornwallis**, in **Bengal, Bihar and Orissa** (and, from **1802**, the **Northern Circars** and parts of Madras); about **19%** of British India. **Zamindars** became **owners** and paid a **fixed sum for ever** (**10/11** to the Company, **1/11** kept). **Sunset Law**: if rent was not paid by sunset on the due date, the estate was **auctioned**. **Effects:** the Company's income became **certain** and a loyal class grew; but **no incentive for improvement**, **cultivators** lost rights, **absentee landlords**, and the Company lost the **rising value of land**.
Ryotwari, 1820	**Captain Alexander Read** began it in **Baramahal (1792)** and **Thomas Munro** extended it as **Collector of the Ceded districts (1800)** and as **Governor of Madras (1820–27)**; in **Madras and Bombay** (about **51%** of British India). The **ryot (cultivator)** is the **owner** and pays **directly** to the State; revenue about **1/3 to 1/2 of produce**, **revised every 20–30 years**. **Effects:** no middleman, but **rates very high**, **heavy debt to moneylenders**, and the State acted like a **landlord**.
Mahalwari, 1822 and 1833	**Holt Mackenzie (1822)**, refined by **Robert Bird (1833)**; in the **North-Western Provinces, Central India and Punjab** (about **30%**). The **mahal (village or group of villages)** is the unit; the **village community or its headmen (lambardars)** pay **jointly**; revenue fixed for **20–30 years**.

4. The four pillars of administration

Pillar	What was done
Civil service	**Cornwallis** gave high salaries and ended private trade; **Fort William College (1800)** and **Haileybury College (1806)** trained recruits; **1853** open competitive exam in London; **Macaulay Committee (1854)**; Indians kept out of **higher posts**. **ICS** name from **1858–61**; first Indian: **Satyendranath Tagore (1863)**. Called the **'steel frame'**.
Army	Mostly **Indian sepoys** under **British officers**; the **Bengal, Madras and Bombay armies**; high posts only for the British; the Company spent most revenue on the **army**. Sepoy discontent later led to **1857**.
Police	**Cornwallis** set up the **district police** under **darogas** and removed the zamindars' police; in the **Madras Presidency**, **Munro** replaced the Palegars' village watch; **Thuggee and Dacoity Department** under Sleeman; the **Police Act 1861** (after 1857) built the modern police.
Judiciary	**Supreme Court, Calcutta (1774)**; **Cornwallis Code (1793)**: Collector no longer a judge, **rule of law**, **government servants answerable to courts**; **Sadr Diwani and Nizamat Adalats**; **Macaulay's Law Commission** gave the **Indian Penal Code (drafted 1837, in force 1860)**; **Indian High Courts Act 1861** set up **High Courts** at Calcutta, Bombay, Madras (**1862**). Faults: **slow and costly justice**, **racial bias** (the **Ilbert Bill** case later).

5. The Andhra link

- **Madras Presidency:** the Telugu country was split among the **Northern Circars** (Visakhapatnam, Godavari, Krishna, Guntur and Srikakulam areas), the **Ceded districts** (Kadapa, Kurnool, Anantapur, Bellary) and

Nellore; Telangana stayed with the **Nizam**. Each region was ruled differently, which is why **zamindari** and **ryotwari** areas existed side by side in Andhra.

- **Northern Circars: Permanent Settlement (1802)** made zamindars (e.g. of **Vizianagaram, Pithapuram, Venkatagiri, Bobbili**) owners. **Ceded districts, Nellore and others: Thomas Munro** introduced **Ryotwari** from **1800**, and ended the **Palegar** system (Palegars lost their forts and police rights).
- **Revolt: Uyyalawada Narasimha Reddy** (1846–47), a former Palegar whose pension was cut, rose against the Company; he attacked the **Koilkuntla treasury** and was **hanged on 22 February 1847**, one of the earliest revolts before 1857.
- **Irrigation: Sir Arthur Cotton** built the **Dowleswaram (Godavari) Anicut (completed 1852)** and the **Krishna Anicut at Vijayawada (1855)**, turning the delta districts into the **'granary of the South'** and bringing the delta **prosperity** (while the Rayalaseema regions stayed dry and poor).

6. Impact of consolidation

Area	Points
Politics	Rule by **law and a central bureaucracy**; **Indian princes** reduced to **dependents** (Subsidiary Alliance, Lapse); **separation** of revenue and justice; **Parliament control** over the Company.
Society	**Sati abolished (1829)**, **Widow Remarriage Act (1856)**, **English education**, but the rulers kept a **racial distance** and Indians were shut out of high posts.
Economy	**High land revenue**, **moneylender debt**, **decline of handicrafts** (British machine cloth), **railways and telegraph** built mainly for **trade and army**; **commercialisation of agriculture** (indigo, cotton, jute).
Religion	Non-interference in principle, yet **missionaries** were allowed (1813) and **social laws** (sati, widow marriage) touched religious custom; fear for religion fed **1857**.

EXAM TRAP

1773 Regulating Act (Hastings, Supreme Court); **1784** Pitt's India Act (Board of Control); **1793** Permanent Settlement, Cornwallis Code; **1813** monopoly ends (tea, China excepted), ₹1 lakh; **1829** sati abolished; **1833** Governor-General of India (Bentinck, Macaulay); **1835** English education; **1848–56** Dalhousie: **Doctrine of Lapse** (Satara 1848, Jhansi 1853, Nagpur 1854), **railway 1853**, **post 1854**, **Wood's Despatch 1854**; **Awadh 1856** (misgovernment); **Ryotwari** (Read, Munro) ~51%; **Mahalwari** ~30%; **Permanent** ~19%; **1802** Circars zamindari; **Cotton's Godavari anicut 1852**; **1846–47** Narasimha Reddy.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements about the Permanent Settlement of 1793:
1. It was introduced by Lord Cornwallis in Bengal, Bihar and Orissa.
2. Zamindars were recognised as owners of the land.
3. The amount payable to the Company was fixed permanently.

4. It was extended to the whole of British India including the Ryotwari areas.

Which of the above are correct?

A. 1 and 2 only

B. 2 and 4 only

C. 1, 2 and 3 only

D. 1, 2, 3 and 4

Answer: C. It covered only about 19% of British India; Madras and Bombay mostly had Ryotwari and the north-west had Mahalwari.

PRACTICE

Match List-I (Act) with List-II (provision):

List-I: A. Regulating Act 1773 B. Pitt's India Act 1784 C. Charter Act 1813 D. Charter Act 1833

List-II: 1. Board of Control 2. Governor-General of Bengal and a Supreme Court at Calcutta 3. Company's monopoly ended except tea and China trade; ₹1 lakh for education 4. Governor-General of India and a Law Member

A. A-1, B-2, C-3, D-4

B. A-2, B-3, C-1, D-4

C. A-2, B-1, C-3, D-4

D. A-2, B-1, C-4, D-3

Answer: C. 1773: Governor-General of Bengal and Supreme Court; 1784: Board of Control; 1813: monopoly ended; 1833: Governor-General of India.

PRACTICE

Which one of the following statements is NOT correct?

A. The Ryotwari system was first introduced by Lord Cornwallis in Bengal.

B. Lord William Bentinck abolished sati by Regulation XVII of 1829.

C. The Wood's Despatch of 1854 led to universities at Calcutta, Bombay and Madras.

D. Under the Cornwallis Code the Collector lost judicial powers.

Answer: A. Ryotwari was associated with Read and Munro in Madras; Cornwallis introduced the Permanent Settlement in Bengal.

PRACTICE

Assertion (A): Lord Dalhousie annexed Satara in 1848 and Jhansi in 1853 under the Doctrine of Lapse.
Reason (R): Awadh was also annexed in 1856 under the Doctrine of Lapse.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: C. A is true; R is false, since Awadh was taken in 1856 on the ground of misgovernment.

PRACTICE

In 1800 an officer was made Collector of the Ceded districts and settled the land revenue directly with each cultivator, without zamindars or middlemen, later extending the system as Governor of Madras. The system he followed is

A. Permanent Settlement

B. Mahalwari system

C. Ryotwari system

D. Subsidiary Alliance

Answer: C. This is the Ryotwari system associated with Read and Thomas Munro.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the six Acts (1773 to 1853) with one change each. 2. Write each Governor-General with two works. 3. Write the three land revenue systems (year, founder, area, who pays) and the Andhra link.

POLITY · Distribution of legislative and executive powers between the Union and States

UNDERSTAND IT FIRST

What is it? India is a **Union of States**: both the **Centre and the States** have their own governments, so the Constitution must say **who may make laws and who may run the administration on each subject**, to prevent clashes. It does this by **dividing subjects into three lists** (Seventh Schedule) and by **linking executive power to legislative power**: the government can **act only where its legislature can legislate**. The Constitution gives the **Centre extra strength**, so India is called **'federal in form, unitary in spirit'** (a **quasi-federal** system); but the **Supreme Court (S.R. Bommai, 1994)** treats **federalism as part of the basic structure**.

The three lists

1. Legislative powers: Arts 245–255

Provision	Details
Art 245	**Territorial limits**: Parliament makes laws for **all or part of India**; a State legislature for **its State**. Only Parliament can make **extra-territorial** laws (e.g. for Indians abroad).
Art 246	**Subject-matter**: Parliament has **exclusive** power on the **Union List**; both can legislate on the **Concurrent List** (Union law prevails); States have **exclusive** power on the **State List** (subject to Parliament's overriding powers). **Union List gets priority** over the other two.
Seventh Schedule	**Union List: 98** subjects (originally 97); **State List: 59** (originally 66); **Concurrent List: 52** (originally 47). The **42nd Amendment (1976)** shifted **five** subjects from State to Concurrent: **education, forests, protection of wild animals and birds, weights and measures, administration of justice**. (Borrowed from the **Government of India Act 1935** and **Canada's** strong Centre, with the **Concurrent List from Australia**.)
Residuary powers (Art 248)	Subjects in **no list** (e.g. a new technology) belong to **Parliament** (Entry 97 of the Union List), unlike the **USA** where they go to the States. **Canada** is the model.
Art 246A (GST)	Inserted by the **101st Amendment (2016)**: both Parliament and States can make laws on **Goods and Services Tax**; the **GST Council** (Art 279A) decides.

2. When the Centre may legislate on State subjects

Article	Situation
Art 249	**National interest**: if the **Rajya Sabha** passes a resolution by **two-thirds of members present and voting** that Parliament should legislate on a **State List** subject; valid for **one year**, extendable by a year each time.
Art 250	**National Emergency**: Parliament can legislate on the State List while the Emergency lasts (laws lapse **six months** after it ends).
Art 252	**Consent of two or more States**: their legislatures pass resolutions asking Parliament to legislate; the law applies to those States (others may **adopt it later**). Example: **Prize Competitions Act**.

Art 253	**International treaties**: Parliament can legislate on any subject to **implement a treaty or international agreement**, even a State List subject.
Art 356	**President's Rule**: when a State's constitutional machinery fails, Parliament exercises the State's legislative powers.
Art 254 and **Art 200–201**	On the **Concurrent List**, if a State law **conflicts (is repugnant)** with a Union law, the **Union law prevails** (254(1)). **Exception:** a State law **reserved for the President and assented to by him** prevails **in that State** (254(2)), though Parliament can later override it. **Governor** may reserve State Bills for the President (Art 200).

3. Doctrines used by the courts

Doctrine	Meaning
Pith and substance	To decide which list a law belongs to, look at its **true nature**; a law mainly on a State subject is valid even if it **incidentally touches** a Union subject (**State of Bombay v F.N. Balsara, 1951**).
Colourable legislation	**'What cannot be done directly cannot be done indirectly'**: a legislature cannot do **indirectly** what it lacks the power to do **directly** (**K.C. Gajapati Narayan Deo v State of Orissa, 1953**).
Harmonious construction	If two entries **overlap**, read them **together** to give effect to both; if impossible, the **Union List prevails**.
Territorial nexus	A State law may apply outside the State if there is a **real connection (nexus)** between the State and the subject (**State of Bombay v R.M.D. Chamarbaugwala, 1957**).

4. Executive powers

Provision	Details
Art 73 (Union)	Union executive power extends to **matters on which Parliament can legislate** and to rights under treaties. **Concurrent List:** it extends only if the **Constitution or a Parliament law** gives it; otherwise the **States** run those laws.
Art 162 (State)	State executive power extends to **matters on which the State legislature can legislate**, and on the Concurrent List it is **subject to Union law**.
Art 256, 257	States must **comply with Union laws**; the Union can give **directions** to States: to keep **national and military communications** safe, to protect **railways**, and (Art 257A was deleted). If a State **disobeys**, **Art 365** lets the President say constitutional machinery has failed, leading to **Art 356**.
Art 258, 258A	The **President** may, with a State's **consent**, entrust **Union functions** to it or its officers; a **Governor** may, with the Centre's consent, entrust **State functions** to the Union.

5. Why the Centre is strong, and the checks

- **Centre's strengths:** longer **Union List**, **residuary powers**, Union law prevails on the Concurrent List, Parliament's power over the State List (Arts 249, 250, 252, 253, 356), **Governor** appointed by the President, **single citizenship**, **all-India services** and a **single integrated judiciary**.
- **Checks and balance:** the **Supreme Court** settles Union–State disputes (**Art 131**); **S.R. Bommai (1994)** made **federalism** a basic feature and made **Art 356** reviewable; **State of West Bengal v Union of India (1963)** held India is **not a true federation** (Union stronger); the **Inter-State Council (Art 263; set up 1990)**; the **Sarkaria Commission (1983–88)** and **Punchhi Commission (2007–10)** studied Centre–State relations.

EXAM TRAP

Art 245 territory; **Art 246** three lists; **Seventh Schedule: Union 98, State 59, Concurrent 52**; **42nd Amendment**: education, forests, wild animals, weights and measures, justice to Concurrent; **Art 248** residuary to Union; **Art 246A** GST (101st Amendment 2016); **Art 249** Rajya Sabha 2/3, one year; **250** Emergency; **252** consent of States; **253** treaties; **356** President's Rule; **Art 254** repugnancy (State law reserved for the President prevails); **Pith and substance** (Balsara 1951); **Colourable** (Gajapati 1953); **Art 73** and **162**; **Art 256–258**; **Art 263** Inter-State Council; **Bommai 1994**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements about the distribution of legislative powers:
1. Residuary powers are vested in Parliament.
2. The 42nd Amendment shifted education and forests from the State List to the Concurrent List.
3. A resolution of the Rajya Sabha by a two-thirds majority of members present and voting under Article 249 enables Parliament to legislate on a State List subject for one year at a time.
4. A State legislature can make laws with extra-territorial operation just like Parliament.
Which of the above are correct?

A. 1, 2 and 3 only

B. 1 and 4 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: A. Only Parliament can make extra-territorial laws (Art 245); State laws are limited to the State, subject to territorial nexus.

PRACTICE

Match List-I (Article) with List-II (subject):
List-I: A. Article 248 B. Article 249 C. Article 252 D. Article 253

List-II: 1. Legislation for two or more States by their consent 2. Residuary powers 3. Legislation to implement international treaties 4. Legislation on a State List subject in the national interest

A. A-2, B-4, C-1, D-3

B. A-4, B-2, C-1, D-3

C. A-2, B-4, C-3, D-1

D. A-2, B-1, C-4, D-3

Answer: A. 248: residuary; 249: national interest; 252: States' consent; 253: treaties.

PRACTICE

Which one of the following statements is NOT correct?

A. On a Concurrent List subject, a Union law generally prevails over a conflicting State law.

B. Parliament can legislate on a State List subject to implement an international treaty.

C. The doctrine of pith and substance is used to decide the true nature of a law when lists overlap.

D. Parliament can never legislate on a State List subject during a National Emergency.

Answer: D. Under Art 250 Parliament can legislate on the State List during a National Emergency.

PRACTICE

Assertion (A): A State law on a Concurrent List subject may prevail over a conflicting Union law within that State.
Reason (R): Under Article 254(2), such a State law prevails if it has been reserved for the consideration of the President and has received his assent.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A. R correctly explains the exception to Union supremacy.

PRACTICE

A State enacts a law that, in its true nature, regulates a State List subject, but it incidentally affects a matter in the Union List. The court upholds it by looking at its dominant character. This is the application of the doctrine of

A. Colourable legislation

B. Pith and substance

C. Territorial nexus

D. Doctrine of eclipse

Answer: B. Pith and substance looks at the true nature of a law; incidental encroachment does not invalidate it.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write Arts 245, 246, 248, 246A and the three lists with numbers and two examples each. 2. Write Arts 249, 250, 252, 253, 356 and Art 254. 3. Write the four doctrines with cases, and Arts 73, 162, 256–258.

MATHS · Time and work

UNDERSTAND IT FIRST

What is it? These problems ask **how long a job takes** when one or more people (or machines or pipes) work on it. The one idea behind all of them: **work done = rate × time**. If a person finishes a job in **n days**, in **one day** the person does **1/n** of the job: that is the **rate**. **More workers means less time**, and **efficient workers do more in the same time**. Treat the **whole job as 1** (or as a convenient number of **units**), then add rates when people work together.

Time and work: key ideas

1. Rate and working together

- **Rate = 1/days.** If **A** takes **a** days and **B** takes **b** days, then together in **one day** they do **1/a + 1/b = (a + b)/(a × b)**, so the **time together = (a × b)/(a + b)** days. **Example:** A = 10 days, B = 15 days: together **150/25 = 6 days**.
- **LCM method (fastest):** take the **LCM of the days** as the **total work in units**. A (10 days) and B (15 days): total = **30 units**; A does **3** units a day, B does **2**; together **5** a day, so **30/5 = 6 days**.
- **Finding one person's time:** if A + B finish in **x** days and A alone in **a** days, B's rate = **1/x − 1/a**. **Efficiency:** if A is **twice as efficient** as B, A takes **half** the time: **rates 2 : 1**, **times 1 : 2**.

- **Leaving midway and alternate days:** work in **stages**: units done in the first stage, then remaining units at the new rate. **Alternate days:** find the work in **two days** (one day each), then count cycles. **Example:** A (10 days) and B (15 days) on alternate days: two days = 3 + 2 = 5 of 30 units, so **12 days** for 30 units.

2. Pipes and cisterns

- It is the **same idea**: an **inlet pipe** fills (a **+** rate), an **outlet pipe** empties (a **−** rate). **Net rate = inlets − outlets.** If the net rate is **negative**, the tank **empties**.
- **Example:** A fills in 20 minutes, B fills in 30, C empties in 15. Take **60 units**: A = +3, B = +2, C = −4 per minute; net **+1** a minute, so **60 minutes**.

3. Wages and the man-days formula

- **Wages are shared in the ratio of work done** (rate × days), not by days alone. **Example:** A (6 days), B (8 days) and C together finish in 3 days. Take **24 units**: A = 4, B = 3 per day, C = 24/3 − 4 − 3 = **1**. Wages ratio **4 : 3 : 1**; on ₹3,200, C gets **₹400**.
- **Man-days formula: $M_1 \times D_1 \times H_1 / W_1 = M_2 \times D_2 \times H_2 / W_2$** (M = men, D = days, H = hours a day, W = work). **Example:** 24 men finish a job in 20 days; after **8 days**, **8 men leave**. Total = 24 × 20 = **480 man-days**; done = 24 × 8 = 192; left = **288**; 16 men need 288/16 = **18 more days**.

EXAM TRAP

Rate = 1/days; together = ab/(a + b); LCM = total units; net rate for pipes = inlets − outlets; wages ∝ work done; $M_1D_1H_1/W_1 = M_2D_2H_2/W_2$; more men → fewer days; if A is k times as efficient as B, A takes 1/k of the time.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

A and B together can complete a work in 12 days, B and C in 15 days, and A and C in 20 days. In how many days can A, B and C together complete the work?

A. 8 days

B. 10 days

C. 12 days

D. 15 days

Answer: B. (A+B) + (B+C) + (A+C) = 1/12 + 1/15 + 1/20 = 12/60 = 1/5, so A+B+C = 1/10: 10 days.

PRACTICE

A is twice as efficient as B, and together they complete a work in 12 days. How many days will A alone take?

A. 16 days

B. 20 days

C. 18 days

D. 24 days

Answer: C. Rates 2 : 1, so together 3 units a day; total = 36 units; A alone = 36/2 = 18 days.

PRACTICE

Pipe A can fill an empty tank in 20 minutes, pipe B in 30 minutes, and pipe C can empty the full tank in 15 minutes. If all three are opened together, in how much time will the empty tank be filled?

A. 60 minutes

B. 45 minutes

C. 30 minutes

D. 90 minutes

Answer: A. LCM 60 units: +3 +2 −4 = +1 per minute, so 60 minutes.

PRACTICE

A can do a work in 6 days and B in 8 days. With the help of C they finish it in 3 days and receive ₹3,200 in all. What is C's share?

A. ₹300

B. ₹400

C. ₹480

D. ₹600

Answer: B. Take 24 units: A = 4, B = 3, C = 1 per day, ratio 4 : 3 : 1; C's share = 3,200/8 = ₹400.

PRACTICE

24 men can finish a job in 20 days. After 8 days, 8 men leave. In how many more days will the remaining men finish the job?

A. 20 days

B. 22 days

C. 16 days

D. 18 days

Answer: D. Total 480 man-days; done 192; left 288; 288/16 = 18 days.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write rate, together, and the LCM method with one example each. 2. Write the pipes rule and the wages rule. 3. Write the man-days formula and solve the 'men leave midway' example.

NOTEBOOK PAGE · Day 14: write this from memory

Close the notes. Write these headings, fill every blank, then correct in a different colour.

HISTORY: Consolidation of British rule

Area	Write
Acts	1773 Regulating, 1784 Pitt's India, 1793, 1813, 1833, 1853 Charter Acts: one change each
Governors-General	Hastings, Cornwallis, Wellesley, Bentinck, Dalhousie: two works each
Land revenue	Permanent, Ryotwari, Mahalwari: year, founder, area, who pays
Pillars	Civil service, army, police, judiciary: reforms and people
Dalhousie	Doctrine of Lapse (states), railways 1853, telegraph, post 1854, Wood's Despatch 1854
Andhra link	Munro (Ryotwari), Circars zamindari 1802, Cotton's anicuts, Palegars, Narasimha Reddy 1846

POLITY: Distribution of powers

- Art 245, 246 (supremacy of Union List), Seventh Schedule (Union 98, State 59, Concurrent 52), residuary Art 248, 246A (GST), 42nd Amendment shifts.
- Parliament over State List: Art 249, 250, 252, 253, 356; Art 254 repugnancy; doctrines: pith and substance, colourable legislation, harmonious construction; executive: Art 73, 162, 256–258.

MATHS: Time and work

Write: rate = 1/days; together = add rates; LCM method; pipes and cisterns (inlet +, outlet −); wages in the ratio of work; $M_1D_1H_1/W_1 = M_2D_2H_2/W_2$; one worked example of each.